

Virtual SOC Environment — Project Overview

Purpose

Build a fully functional, cloud-hosted Security Operations Center (SOC) environment for cybersecurity training, research, and demonstration. The environment must simulate realistic adversary activity, provide full network visibility, support intrusion detection, log aggregation, endpoint monitoring, and enable blue-team/incident-response workflows.

Objectives

- Deploy a multi-segment virtual network inside OpenStack that mirrors a production SOC architecture.
- Implement pfSense as the central firewall/router handling segmentation, NAT, and routing.
- Deploy Suricata as the Network Intrusion Detection System (NIDS).
- Deploy Wazuh as SIEM + EDR for centralized alerting, correlation, endpoint telemetry, and dashboards.
- Integrate pfSense, Suricata, Wazuh agents, and remote nodes for full-pipeline logging.
- Implement WireGuard VPN for secure operator access.
- Deploy attacker and victim hosts (Kali, Metasploitable 3) for controlled offensive testing.
- Support demonstrations of scanning, malware activity, detections, and investigation workflows.

High-Level Architecture

- **ext_net → OpenStack Neutron Router → transit-net (/29) → pfSense WAN**
- **pfSense LAN → internal network (/24) hosting:**
 - Victim VM(s)
 - Metasploitable 3
 - NIDS visibility domain
 - Wazuh Agents
- **pfSense WAN + WireGuard** provides:
 - Remote access for SOC operators
 - Bastion SSH pipeline

Major Components

pfSense Firewall

- WAN: transit-net
- LAN: internal 105.45.5.0/24
- Provides NAT, routing, firewalling, DHCP, SSH bastion, WireGuard termination.

Suricata NIDS

- Installed on pfSense

- Operates on LAN interface
- Uses ET Open ruleset
- Captures and forwards EVE JSON logs

Wazuh SIEM + EDR

- Installed on Ubuntu 22.04
- Provides log indexing, rule correlation, dashboards, alerting
- Receives:
 - Syslog-ng forwarded Suricata alerts
 - pfSense logs
 - Agent telemetry from endpoints

Endpoints & Attack Hosts

- Victim hosts for telemetry generation
- Kali attacker for port scans, exploitation, and rule verification
- Metasploitable 3 for realistic high-noise attack surface

Core Capabilities of the Environment

- Network telemetry collection via Suricata
- Host telemetry via Wazuh agents
- Log centralization and correlation via Wazuh Indexer
- Dashboard visualization of:
 - Attacks
 - Scans
 - Malware signatures
 - System anomalies
 - Secure remote operator access through WireGuard
 - Controlled offensive operations from Kali via routed attack paths

Deliverables

- Fully configured multi-segment VSOC environment
- Documented installation and configuration steps
- Operational demo showcasing:
 - Suricata NIDS detections
 - Wazuh SIEM correlation
 - Agent-based host alerts
 - Attack chains using Nmap, Metasploit, etc.
 - Pipeline for future research, student labs, and continuous improvement.

Dependencies

- OpenStack tenant access
- Ubuntu 22.04 LTS for Wazuh
- pfSense 2.8.1

- Suricata package
- syslog-ng package
- WireGuard package
- Kali Linux image
- Metasploitable 3 QCOW2 image

Status Summary

The architecture, networking, SIEM base, NIDS, firewall routing, and initial telemetry pathways have been fully validated. Forward steps include: - Deep tuning of detection rules - Expanded endpoint coverage - Additional lab scenarios and attacker workflows - Visualization dashboards for instructional use.